

Automated & Emerging Technologies

Syllabus

6.1 Automated Systems

- 1 Describe how sensors, microprocessors and actuators can be used in collaboration to create automated systems
- 2 Describe the advantages and disadvantages of an automated system used for a given scenario
 - Including scenarios:
 - Industry
 - Transport
 - Agriculture
 - Weather
 - Gaming
 - Lighting
 - Science

6.2 Robotics

- 1 Understand what is meant by Robotics
 - Robotics is a branch of computer science that incorporates the design, construction and operation of robots.
 - Examples include factory equipment, domestic robots and drones.
- 2 Describe the characteristics of Robot
 - Including:
 - A mechanical structure or framework
 - Electrical components, such as sensors, microprocessors and actuators
 - Programmable
- 3 Understand the roles that robots can perform and describe the advantages and disadvantages of their use.
 - Robots can be used in areas including:
 - Industry
 - Transport
 - Agriculture
 - Medicine
 - Domestic
 - Entertainment



6.3 Artificial Intelligence

- 1 Understand what is meant by artificial intelligence (AI).
 - AI is a branch of computer science dealing with the simulation of intelligent behaviors by computers.
- 2 Describe the main characteristics of AI as the collection of data and the rules for using that data, the ability to reason, and can include the ability to learn and adapt
- 3 Explain the basic operation and components of AI systems to simulate intelligent behavior.
 - Limited to:
 - Expert Systems
 - Machine Learning
 - Expert systems have a knowledge base, a rule base, an inference engine and an interface.
 - Machine learning is when a program has the ability to automatically adapt its own processes and/or data

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6.1 Automated Systems

Sensors, Microprocessors & Actuators

Automated System, combination of hardware & software, designed and programmed to work automatically without any human intervention. Monitoring is required.

- Input devices which take readings from surroundings and send this data to a microprocessor or computer.
- Microprocessor will process the data and take action accordingly.
- Form of output, usually involving signals sent to actuators to control motors, wheels, solenoids, etc.

Role of Automated Systems

Role of Microprocessors or Actuators in the following Application areas:

Industrial Application

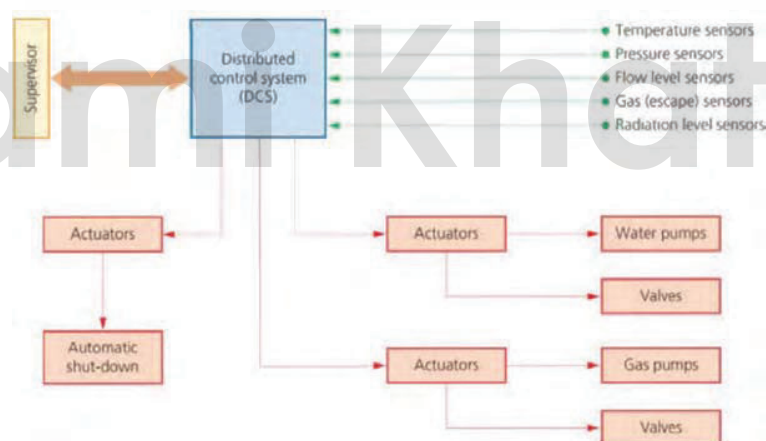
Quality and Flexibility has increased with increase in automation.

Example: Manufacture of car engines, when done manually, the installation of pistons into the engine had an error-rate of -1.5%; with automated systems, the error-rate has fallen to 0.00001%.

Two major Industrial Applications are:

Nuclear Power Station

Automation gives increased safety both in the process itself and to the workforce. Centre of the system is a distributed control system (DCS). DCS is essentially a powerful computer that has been programmed to monitor and control the whole process with no human interaction required.



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Pros

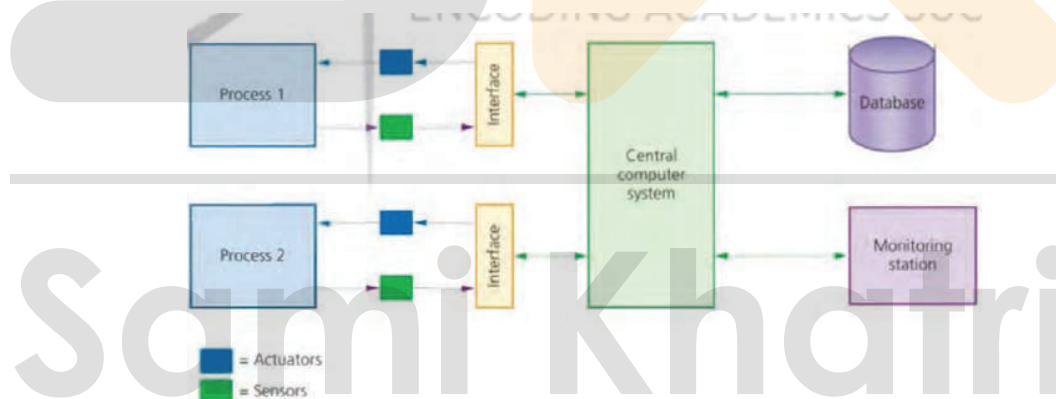
- Faster
- Safer
- Cheap
- Run under optimum conditions

Cons

- Expensive
- Safety Implications
- Cyberattacks
- Enhanced Maintenance

Manufacture of Paracetamol

Automated System also depends on sensors, a computer, actuators and software. Process 1 is the manufacture of the paracetamol. Process 2 is the making of the solid tablets. Both processes are monitored by a number of sensors that send their data back to a central computer. It is fully automated, but the operator can override the central computer system if necessary.



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Pros

- Faster
- Safer
- Less expensive
- Run under optimum conditions
- Efficient use of materials
- Higher productivity
- Consistent results

Cons

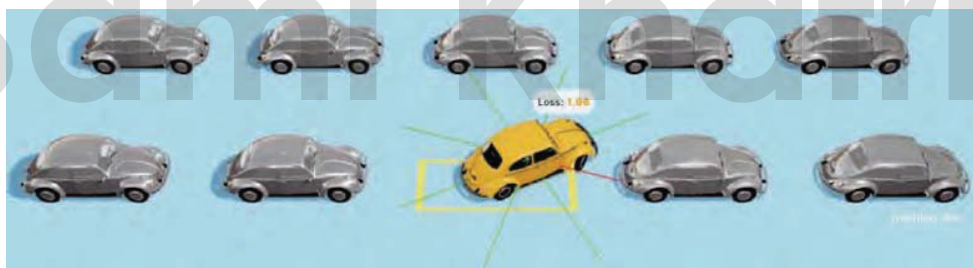
- Expensive
- Safety Implications
- Cyberattacks
- Enhanced Maintenance

Transport

As with industrial processes, many of the automated systems in transport refer to robotic systems (for example, autonomous buses/cars, autonomous trains and unpiloted aircraft).

Automated systems are manually controlled transport, which includes cars, buses/lorries, trains and aircraft.

The driver goes along the row of parked cars. On-board sensors and cameras gauge the size of any parking spaces, and the on-board computer warns the driver if a suitable space has been found. The driver then selects auto-parking and the on-board computer takes over. Actuators are used to operate the steering rack, brakes and throttle under the full control of the computer. This allows the car to go from Step 1 to Step 2 automatically and complete the parking maneuver with no driver intervention.



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Pros

- Use Fewer Parking Space
- Avoid Traffic Disruption
- Cars can fit into smaller spaces
- Reduced insurance claims
- Safer system
- Consistent results

Cons

- Loss of skills
- Malfunction
- Kerbing of wheels
- Expensive
- Enhanced Maintenance

Adaptive Cruise Control

Adaptive cruise control makes use of sensors, an on-board computer and actuators to allow a car to remain a safe distance from another vehicle.

Agriculture

Consider one important example that is being used in Brazil to irrigate crops automatically

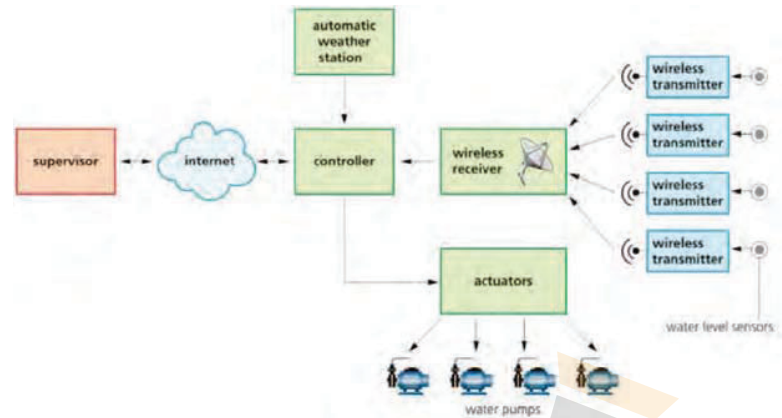
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The watering of crops (irrigation) is fully automatic and also involves considerable amounts of wireless transmission. This allows the system to be used in very remote areas that are vast in size - some of the fields are more than 10 km² in area.

Pros

- Reduced labor
- Better & More Efficient Control
- Faster Response
- Safer
- Different crops may require different irrigation requirements

Cons

- Expensive
- High maintenance costs
- Need to maintain water channels

Weather

Automated weather stations are designed to save labor and to gather information from remote regions or where constant weather data is a requirement. Automated weather stations require a microprocessor, storage (database), battery (usually with solar-powered charging) and a range of sensors:

- Thermometer (to measure temperature)
- Anemometer (to measure wind speed)
- Hygrometer (to measure humidity)
- Barometer (to measure rain fall)
- Light Sensor (to measure hours of daylight)

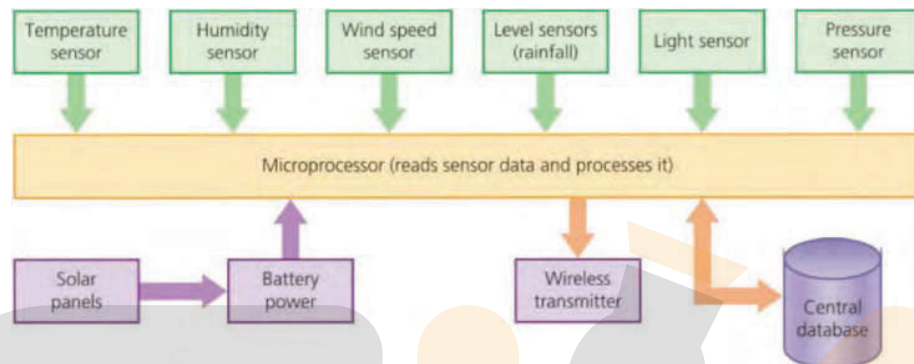


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The data from the sensors and the calculated values are then stored on a central database. Some automated weather stations are sited near airports, where reports are sent out automatically every five minutes to pilots in the vicinity of the airport.



Gaming

Gaming devices involve sensors to give a degree of realism to games:

- Accelerometers
- proximity sensors

Embedded accelerometers and proximity sensors in games consoles allow increased human interaction with the game. This allows players to take actions that simulate real events happening, giving a more immersive games experience.

Lightening

Control of lighting in an office using:

- Light sensors
- Motion sensors
- Infrared sensors

Lightening System in House

The example we will consider here is used in a house:

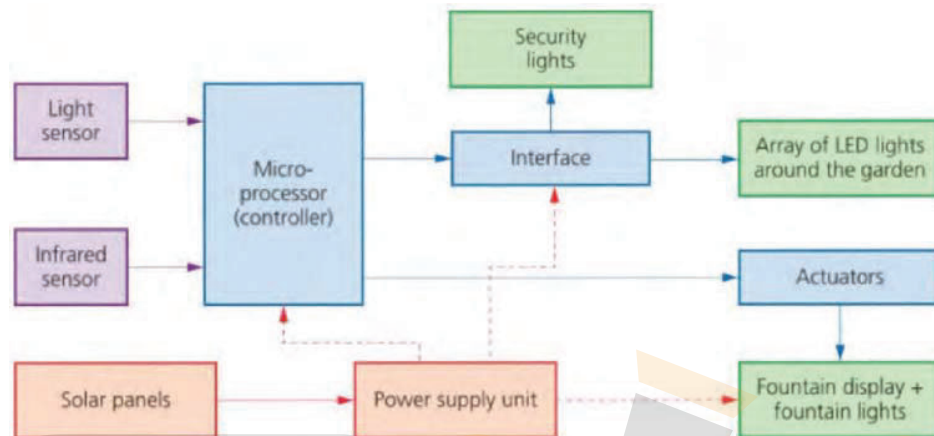
- Where lights in the garden are turned on automatically when someone enters the garden or it turns dark
- Where a lighting show is part of a microprocessor-controlled water fountain display; the lighting only comes on when it becomes dark.



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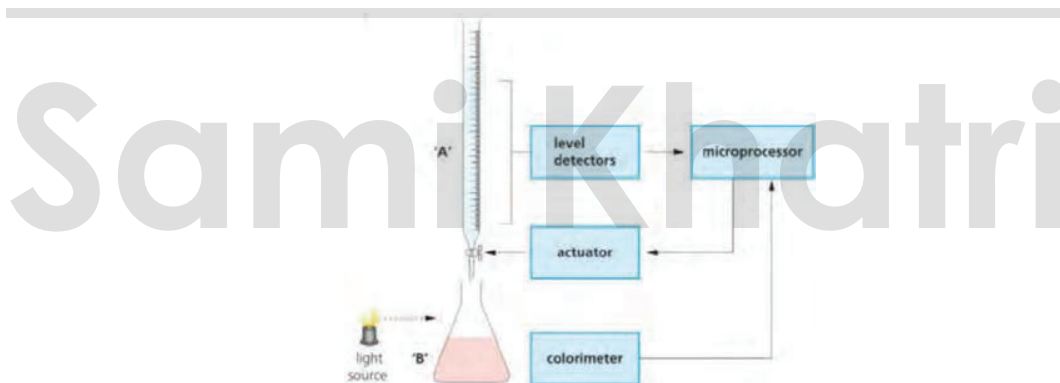
As it becomes dark, the light sensor value will change, and the microprocessor will send signals to the interface to control the array of LED lights around the garden. Data from the infrared sensor would also be used as a security device whenever the house is unoccupied.

Science

Automated systems in scientific research are widely used. There are literally thousands of possible applications. The example we will use here is the automatic control of a laboratory experiment which requires accuracy and repeatability.

Chemical Process in a Laboratory

Imagine an experiment in a pharmaceutical laboratory where two chemicals are reacted together in a vessel. One of the chemicals is being added from a piece of equipment known as a burette to a reaction vessel.



The level sensors measure how much liquid is being added from A; this data is sent to a microprocessor. Readings are also sent to the microprocessor from a colorimeter next to vessel B. The microprocessor controls the opening and closing of the tap in A; this is done by sending signals to an actuator that operates the tap.



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Pros

- More consistent
- Less dangerous
- Faster Results
- Automatic Analysis
- Highly trained staff
- Results can be monitored

Cons

- Less flexible
- Security risks
- Expensive

Finally, there are many automated systems being used in both industry and scientific research that incorporate artificial intelligence (AI). It is therefore worth considering the generic advantages of using AI in these automated systems

Pros

- Ability to store & access vast amount of facts
- Learn from huge amount of available data
- Able to see patterns in results

Cons

- Less flexible
- Security risks
- Expensive



6.2 Robotics

What are Robotics?

Concept of the robot has fired the imagination of science fiction writers for countless years; indeed Isaac Asimov even composed his Three Laws of Robotics:

- Robot may not injure a human through action or inaction
- Robot must obey orders given by humans, unless it comes into conflict with law 1
- a robot must protect itself, unless this conflicts with law 1.

So, what is a robot in the real world? Robotics is a branch of (computer) science that brings together the design, construction and operation of robots. Robots can be found in:



Factories

- Welding parts
- Sprays painting panels on a car
- Fitting windscreens to cars
- Cutting out metal parts to a high precision
- Bottling and Labelling plants
- Warehouses



Home

- Autonomous floor sweepers
- Autonomous lawn mowers
- Ironing robots
- Pool cleaning
- Automatic window cleaner
- Entertainment



Drones

- UAV's are drones that are either controlled remotely or totally autonomous using embedded systems
- Can be used in reconnaissance
- Can be used to make parcel deliveries



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Characteristics of Robot

- 1 Ability to sense their surroundings
- 2 Degree of movement
- 3 Programmable

It is important to realize that robotics and artificial intelligence (AI) are almost two entirely different fields:

Robots don't possess artificial intelligence (AI) since they tend to do repetitive tasks rather than requiring adaptive human characteristics.

It is important not to confuse physical robots with software robots.

Physical robots can be classified as independent or dependent:

- 1 Independent Robots
 - ➔ No direct human control
 - ➔ Replace the human activity totally
- 2 Dependent Robots
 - ➔ Human who is interfacing directly with the robot
 - ➔ Supplement, rather than totally replace, the human activity

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6.3 Artificial Intelligence (AI)

Introduction

Branch of Computer Science dealing with simulation of intelligent human behavior by computer. Known as the cognitive function of human brain. cognitive functions can be replicated in a machine, and they can be measured against human benchmarks such as reasoning, speech and sight.

Characteristics of AI

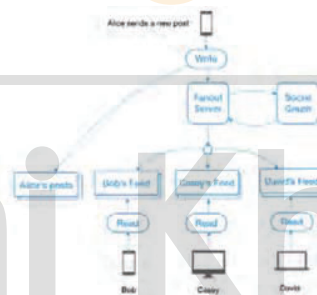
AI is really just a collection of rules and data, and the ability to reason, learn and adapt to external stimuli. AI can be split into three categories:

- ➔ **Narrow AI:** this occurs when a machine has superior performance to a human when doing one specific task
- ➔ **General AI:** this occurs when a machine is similar in its performance to a human doing a specific task
- ➔ **Strong AI:** this occurs when a machine has superior performance to a human in many tasks.

Reasoning is the ability to draw reasoned conclusions based on given data/ situations. Deductive reasoning is where a number of correct facts are built up to form a set of rules which can then be applied to other problems.

Examples

- ➔ News generation based on live news feeds



- ➔ Smart home devices



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➤ Chatbots



➤ Autonomous Cars



➤ Facial Expression Recognition



AI Systems

Two types of AI Systems

- Expert Systems
- Machine Learning

Expert Systems

Computer System that mimics the decision-making ability of a human; expert systems use AI to simulate the judgement and behavior of a human or organization that has expert knowledge and experience.

Expert System will supply a conclusion and any suggested actions to take and it will also give the percentage probability of the accuracy of its conclusions.



User Interface

- Expert Systems interact with user
- Interaction can be through dialogue boxes, command prompts or other input methods



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- Questions being asked usually only have Yes/No answers and are based on the responses to previous questions.

Interface Engine

- Main processing element.
- Acts like a search engine examining the knowledge base for information/data that matches the queries.
- Responsible for gathering information from the user by asking a series of questions and applying responses where necessary.
- It is the problem-solving part of the expert system that makes use of inference rules in the rules base.
- Knowledge Base is a collection of objects and attributes.

Knowledge Base

- Knowledge Base is a repository of facts.
- Stores all knowledge about an area of expertise from a number of expert resources.
- Collection of objects & their attributes.

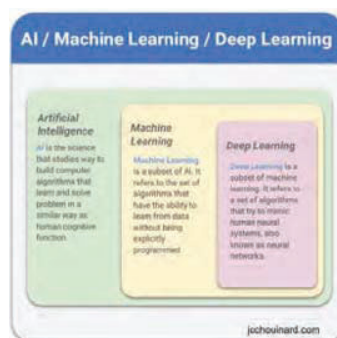
Rules Base

- Set of inference rules
- Inference rules are used by the inference engine to draw conclusions
- Follow logical thinking

Setting Expert System

- Data gathered from human experts or written sources.
- Gathered Data is used to populate the knowledge base that needs to be first created
- Rules base needs to be created
- Inference engine itself needs to be set up
- User interface needs to be developed to allow the user and the expert system to communicate
- System is setup, it needs to be fully tested.

Machine Learning

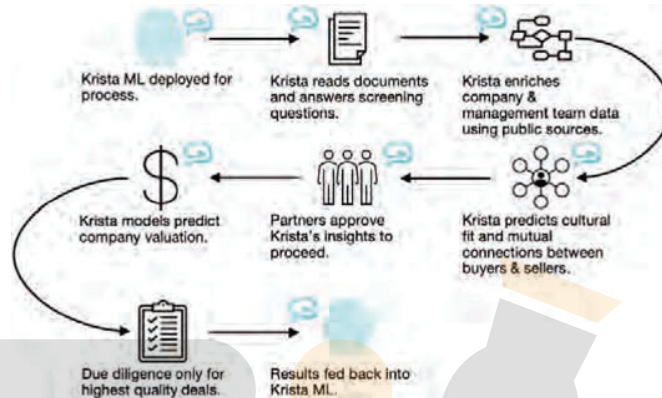


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Science of training computers with sample data so that they can go on to make predictions about new unseen data, without the need to specifically program them for the new data.



Artificial Intelligence

Machine Learning



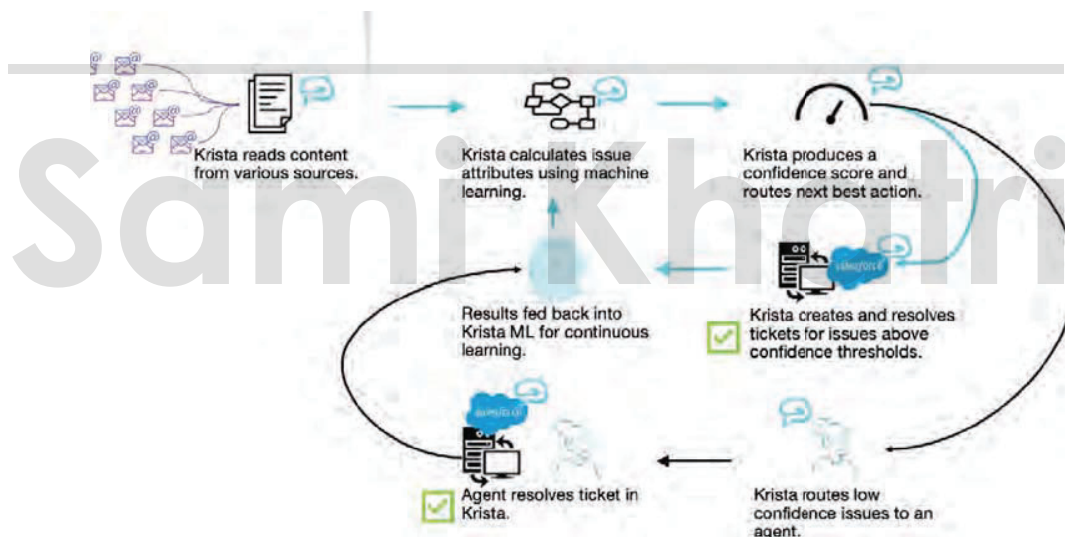
☐ Represents simulated intelligence in machines.

☐ Decision making ability without being programmed

☐ Build machines that are capable of thinking like humans.

☐ Machine learn through data acquisition.

Examples



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